

Addressing Unmet Need in T2DM Through Triple Drug Fixed-Dose Combination of Sitagliptin + Glimepiride + Metformin: Rationale for Combination

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Background

Prevalence of T2DM in India¹

2019:
77 million



2045:
Expected to rise by
134 million

T2DM in Asian Indian people is characterized by:²



Young age of onset



Low levels of BMI

Rationale for combination

- Only less than 50% patients with T2DM achieve their recommended glycemic goals³
- Among Indian Patients with diabetes, mean HbA1c is 8.9%⁴
- Possible reasons for the same include³
 - High pill burden and complex treatment regimens decrease adherence to therapies³
 - Therapy that causes adverse events (e.g.: weight gain, CV problems, hypoglycemia)³
- Each 10% increase in compliance is associated with a 0.1% HbA1c reduction⁵

What are guideline recommendations for management of T2DM?

2020: RSSDI-ESI recommendations⁶

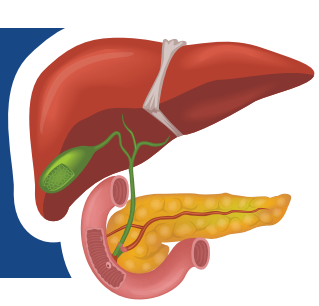
- Consider initiating combination therapy if HbA1c > 1.5 above the target initial therapy

2022: AACE recommendations⁷

- For newly diagnosed persons with T2DM and an entry A1C > 9% and/or ≥ 1.5% above target, one should initiate, along with lifestyle modifications, dual- or possibly triple-combination pharmacotherapy usually including metformin

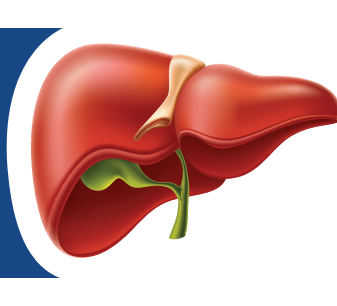
Sitagliptin, metformin and glimepiride in T2DM

Sitagliptin



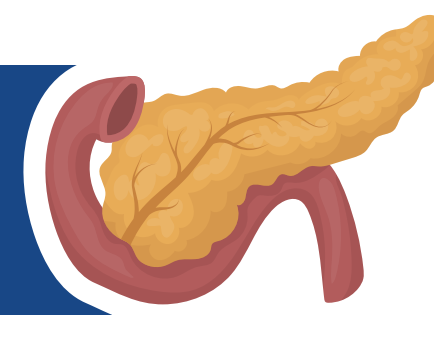
- Inhibits DPP-4 enzyme and thereby prevents degradation of GLP-1⁸
- This causes an increase in insulin release after meals and improved glucose tolerance⁸
- Demonstrated better response in Asians⁹

Metformin



- Suppresses hepatic glucose output, reduction of insulin resistance and increase in GLP-1 release¹⁰
- First-line therapy as per all guidelines^{6,7,11}

Glimepiride



- Stimulate insulin release from pancreatic beta cells¹²
- Very potent and commonly used in India to achieve glycemic targets¹³

Addition of sitagliptin to metformin and glimepiride improves glycemic control and insulin secretion capacity in T2DM

Multinational, randomized, double-blind, parallel-group study (n=194)¹⁴
Addition of sitagliptin to glimepiride + metformin reduced HbA1c by 0.89%

Prospective, 52-week, single centre, single arm, intervention study (n=39)¹⁵
Addition of sitagliptin to glimepiride with/without metformin reduced HbA1c by 0.81%, p<0.001

• Further, addition of sitagliptin to combination therapy:¹⁵

- Reduces dose of glimepiride and preserves β -cell function
- Reduces incidence of hypoglycemia

Safety of sitagliptin + metformin + glimepiride in T2DM

Sitagliptin + metformin + glimepiride^{16,17}



Metformin
Sitagliptin
Glimepiride

- Well-tolerated
- Weight neutral

→ Proven CV safety¹⁸

→ Proven CV safety through large TECOS study¹⁹

→ Proven CV safety²⁰

Sitagliptin + metformin + glimepiride: Impact across value chain of T2DM management

Patient impact^{13,14,15}

- Better HbA1c control
- Improves β -cell function
- Low risk of hypoglycaemia
- Improves compliance

Healthcare system impact²¹

- Decrease in healthcare resource utilization and strain on healthcare
- Reduces/delays long-term complications

Community impact²¹

- Decrease in economic impact
- Improves quality of life

For chronic conditions such as T2DM, a once-daily dosing option has the potential to optimize compliance, and may enhance patient convenience and ease of use which may translate into better treatment outcomes

Conclusion

- Sitagliptin + glimepiride + metformin in fixed-dose combination has the potential to provide better glycaemic control without increased risk of hypoglycaemia, improve treatment compliance and help patients to achieve their glycaemic targets, thereby reducing long-term complications

T2DM: Type 2 diabetes mellitus; BMI: Body mass index; CV: Cardiovascular; HbA1c: Glycated hemoglobin; RSSDI-ESI: Research Society for the Study of Diabetes in India-Endocrine Society of India; AACE: American Association of Clinical Endocrinologists; GLP-1: Glucagon-like peptide 1; DPP-4: Dipeptidyl peptidase-4; TECOS: Trial Evaluating Cardiovascular Outcomes with Sitagliptin

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